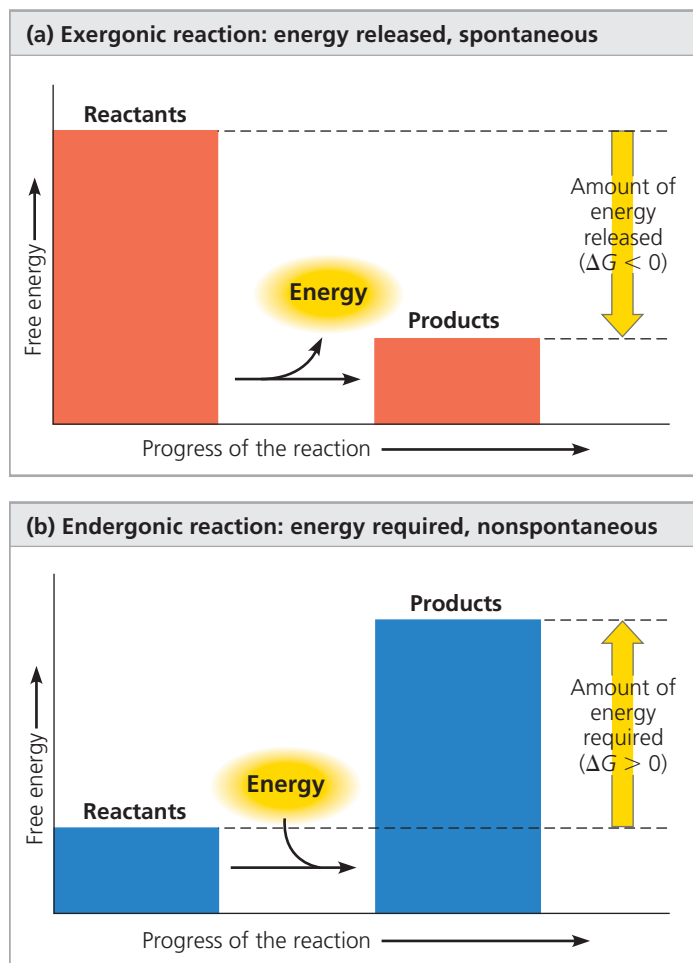


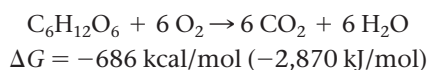
▼ **Figure 8.6** Free energy changes (ΔG) in exergonic and endergonic reactions.



➔ **Mastering Biology Animation: Exergonic and Endergonic Reactions**

represents the maximum amount of work the reaction can perform.* The greater the decrease in free energy, the greater the amount of work that can be done.

We can use the overall reaction for cellular respiration as an example:



For each mole (180 g) of glucose broken down by respiration under what are called “standard conditions” (1 M of each reactant and product, 25°C, pH 7), 686 kcal (2,870 kJ) of energy is made available for work. Because energy must be conserved, the chemical *products* of respiration store 686 kcal less free energy per mole than the *reactants*. The products are,

*The word *maximum* qualifies this statement because some of the free energy is released as heat and cannot do work. Therefore, ΔG represents a theoretical upper limit of available energy.

in a sense, the spent exhaust of a process that tapped the free energy stored in the bonds of the sugar molecules.

It is important to realize that the breaking of bonds does not release energy; on the contrary, as you will soon see, it requires energy. The phrase “energy stored in bonds” is shorthand for the potential energy that can be released when new bonds are formed after the original bonds break, as long as the products are of lower free energy than the reactants.

An **endergonic reaction** is one that absorbs free energy from its surroundings (**Figure 8.6b**). Because this kind of reaction essentially *stores* free energy in molecules (G increases), ΔG is positive. Such reactions are non-spontaneous, and the magnitude of ΔG is the quantity of energy required to drive the reaction. If a chemical process is exergonic (downhill), releasing energy in one direction, then the reverse process must be endergonic (uphill), using energy. A reversible process cannot be downhill in both directions. If $\Delta G = -686 \text{ kcal/mol}$ for respiration, which converts glucose and oxygen to CO_2 and H_2O , then the reverse process—the conversion of CO_2 and H_2O to glucose and oxygen (O_2)—must be strongly endergonic, with $\Delta G = +686 \text{ kcal/mol}$. Such a reaction would never happen by itself.

How, then, do plants make sugar? They get the required energy (686 kcal to make a mole of glucose) by capturing light from the sun and converting its energy to chemical energy. Next, in a long series of exergonic steps, they gradually spend that chemical energy to assemble glucose molecules.

Equilibrium and Metabolism

Reactions in an isolated system eventually reach equilibrium and can then do no work, as illustrated by the isolated hydroelectric system in **Figure 8.7**. The chemical reactions of metabolism are reversible, and they, too, would reach equilibrium if they occurred in the isolation of a test tube. Because systems at equilibrium are at a minimum of G and can do no work, a cell that has reached metabolic equilibrium is dead! *The fact that metabolism as a whole is never at equilibrium is one of the defining features of life.*

▼ **Figure 8.7** Equilibrium and work in an isolated hydroelectric system. Water flowing downhill turns a turbine that drives a generator providing electricity to a lightbulb, but only until the system reaches equilibrium.

